

A First Course in Causal Inference Ch 6

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Ch. 6 Randomization and Regression Adjustment

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Motivation

- ▶ Stratification and post-stratification is for design and analysis with discrete covariates.
- ▶ How should we deal with multidimensional, possibly continuous covariates?

	design	analysis
discrete covariate	stratification	analysis
general covariate	rerandomization	regression adjustment

6.1 Rerandomization

- ▶ Setup: a finite population of n units; n_1 units in a treatment group; n_0 in a control group; $\mathbf{Z} = (Z_i, \dots, Z_n)$ is the treatment vector for these units; and unit i has covariate $X_i \in \mathbb{R}^{K^1}$.
- ▶ Stack them as an $n \times K$ covariate matrix $\mathbf{X} = (X_1, \dots, X_n)^T$ and center them as mean zero $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = 0$.

¹This covariate X_i can have continuous or binary components

6.1 Rerandomization

- ▶ In a CRE, the covariates in the treatment and control groups are balanced **on average** (*not for every stratum!*). Namely, let

$$\hat{\tau}_X = \frac{1}{n_1} \sum_{i=1}^n Z_i X_i - \frac{1}{n_0} \sum_{i=1}^n (1 - Z_i) X_i,$$

then $\mathbb{E}[\hat{\tau}_X] = 0$.

- ▶ In the realized treatment allocation, $\hat{\tau}_X$ is often not zero.
- ▶ Building on Neyman (1923),

$$\text{cov}(\hat{\tau}_X) = \frac{S_X^2}{n_1} + \frac{S_X^2}{n_0} = \frac{n}{n_1 n_0} S_X^2, \quad (1)$$

where $S_X^2 = \frac{1}{n_1} \sum_{i=1}^n X_i X_i^T$ is the finite-population covariance matrix of the covariates.

6.1 Rerandomization

Definition 6.0 Mahalanobis Distance

Given a probability distribution $Q \in \mathbb{R}^N$, with mean $\boldsymbol{\mu} = (\mu_1, \dots, \mu_N)^T$ and positive semi-definite covariance matrix $\boldsymbol{\Sigma}$, the Mahalanobis distance of a point $\mathbf{x} = (x_1, \dots, x_N)^T$ from Q is defined as:

$$M(\mathbf{x}, Q) = \sqrt{(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})}.$$

Let $\mathbf{y} = (y_1, \dots, y_N)^T$ be another point, then the Mahalanobis distance between $\mathbf{x}$ and $\mathbf{y}$ is defined as:

$$M(\mathbf{x}, \mathbf{y}) = \sqrt{(\mathbf{x} - \mathbf{y})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \mathbf{y})}.$$

6.1 Rerandomization

Following Def. 6, the difference between the treatment and control groups is

$$M = \hat{\tau}_X^T \text{cov}(\hat{\tau}_X)^{-1} \hat{\tau}_X = \hat{\tau}_X^T \left(\frac{n}{n_1 n_0} S_X^2 \right)^{-1} \hat{\tau}_X.$$

Lemma 6.1

M remains the same if we transform X_i to $b_0 + BX_i$ for all units $i = 1, \dots, n$ where $b_0 \in \mathbb{R}^K$ and $B \in \mathbb{R}^{K \times K}$ is invertible.

► Proof

- With large n , the Mahalanobis distance M is approximately χ_K^2 under the CRE.
- Rerandomization avoids covariate imbalance by discarding the treatment allocations with large values of M .

6.1 Rerandomization

Definition 6.1 ReM(Rerandomization)

Draw $\mathbf{Z}$ from CRE and accept it if and only if

$$M \leq a,$$

for some predetermined constant $a > 0$.

- ▶ To put it simply, rerandomization keeps randomizing until the treatment and control groups are **close enough**, $M \leq a$.
- ▶ When $a = \infty$, we just conduct a CRE.
- ▶ When $a = 0$, there are very few feasible treatment allocations, then consequently, the experiment has little randomness.
- ▶ In practice, we choose a small but not too extreme a or some upper quantile of a χ_K^2 distribution.

6.1 Rerandomization

Condition 6.1

As $n \rightarrow \infty$,

1. $\frac{n_1}{n}$ and $\frac{n_0}{n}$ have positive limits;
2. the finite population covariance of $\{X_i, Y_i(1), Y_i(0), \tau_i\}$ has a finite limit;
3. $\max_{1 \leq i \leq n} \frac{\{Y_i(1) - \bar{Y}(1)\}^2}{n} \rightarrow 0$, $\max_{1 \leq i \leq n} \frac{\{Y_i(0) - \bar{Y}(0)\}^2}{n} \rightarrow 0$ and $\max_{1 \leq i \leq n} \frac{X_i^T X_i}{n} \rightarrow 0$.

6.1 Rerandomization

Intuitive understanding for Condition 9:

1. With large n , neither of n_1 or n_0 is not extremely small.
2. We exclude a sample such that e.g., the variance of $Y_i(1)$ diverges to ∞ as $n \rightarrow \infty$.
3. One unit does not dominate the sample.

6.1 Rerandomization

Let

$$L_{K,a} \sim D_1 \mid \mathbf{D}^T \mathbf{D} \leq a,$$

where $\mathbf{D} = (D_1, \dots, D_K)$ follows a K -dimensional standard Normal distribution; let ε follow a univariate standard Normal distribution; and $L_{K,a} \perp \varepsilon$.

Theorem 6.1

Under ReM with $M \leq a$ and Condition 9, we have

$$\hat{\tau} - \tau \sim \sqrt{\text{var}(\hat{\tau})} \left\{ \sqrt{R^2} L_{K,a} + \sqrt{1 - R^2} \varepsilon \right\},$$

where $\text{var}(\hat{\tau}) = \frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n}$ is a Neyman (1923)'s variance formula and $R^2 = \text{corr}^2(\hat{\tau}, \hat{\tau}_X)$ is the squared multiple correlation coefficient between $\hat{\tau}$ and $\hat{\tau}_X$ under the CRE.

6.1 Rerandomization

Interpretation of Thm. 6.1

- ▶ $\hat{\tau}$ decomposes into a component that is a linear combination of $\hat{\tau}_X$ and a component that is orthogonal to $\hat{\tau}_X$.
 - ▶ Suppose that X is pre-test score and that the higher the pre-test score is the higher the post-test score. In this case, $\hat{\tau} - \tau$ depends on the covariate imbalance.
 - ▶ Even if $\hat{\tau}_X = 0$, meaning that the covariate is completely balanced, $\hat{\tau}_X - \tau$ depends on the other factors.
- ▶ ReM mitigates the covariate imbalance, so...
 - ▶ When R^2 is large (huge covariance imbalance), ReM successfully decrease $\hat{\tau} - \tau$.
 - ▶ When R^2 is small (small covariance imbalance), ReM does not decrease $\hat{\tau} - \tau$ so much.
- ▶ $\rightsquigarrow$ The efficiency gain due to ReM depends on R^2

6.1 Rerandomization

Proposition 6.1

Under the CRE, we have

$$R^2 = \text{corr}^2(\hat{\tau}, \hat{\tau}_X) = \frac{\frac{S^2(1 | X)}{n_1} + \frac{S^2(0 | X)}{n_0} - \frac{S^2(\tau | X)}{n}}{\frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n}},$$

where $\{S^2(1), S^2(0), S^2(\tau)\}$ are the finite population variances of $\{Y_i(1), Y_i(0), \tau_i\}_{i=1}^n$, and $\{S^2(1 | x), S^2(0 | x), S^2(\tau | x)\}$ are the corresponding finite population variances of their linear projections on $(1, X_i)$.

6.1 Rerandomization

- ▶ When $0 < a < \infty$ the asymptotic distribution of $\hat{\tau}$ has a more complicated form and is more concentrated at τ .
- ▶ Thus $\hat{\tau}$ is more precise under ReM than a CRE.
- ▶ We can still use Neyman (1923)'s conservative CIs but it is overly conservative.

6.2 Regression adjustment

- ▶ In a case that we do not conduct rerandomization in a design stage but we want to adjust the covariate imbalance in the analysis stage, we practice several *regression adjustment* strategies.
- ▶ Covariate-adjusted FRT: the covariates $\mathbf{X}$ are fixed and under H_{0F} the observed outcomes are all fixed.
 - ▶ Recall H_{0F} : $Y_i(1) = Y_i(0)$ for $i = 1, \dots, n$.
- ▶ Here we can compute the test statistic $T = T(\mathbf{Z}, \mathbf{Y}, \mathbf{X})$.
- ▶ There are two general strategies to construct T .

6.2 Regression adjustment

Definition 6.2 (pseudo-outcome strategy for covariate-adjusted FRT)

We can construct the test statistic based on the residuals from fitted statistical models. We can regression Y_i on X_i to obtain residuals $\hat{\varepsilon}_i$ and then treat $\hat{\varepsilon}_i$ as the pseudo outcome to construct test statistics.

Definition 6.3 (model-output strategy for covariate-adjusted FRT)

We can use a regression coefficient as a test statistic. We can regress Y_i on (Z_i, X_i) to obtain the coefficient of Z_i as the test statistic.

- ▶ Def. 6.2: We want to ensure that the pseudo-outcome $(\hat{\tau}_i)$ satisfies H_{0F} if the original outcome Y_i satisfies H_{0F} .
- ▶ Def. 6.3: We want to use the coefficient of Z_i to measure the deviation from H_{0F} .

6.2 Regression adjustment

- ▶ Direct estimation of the average causal effect τ that adjusts for the observed covariates.
- ▶ Fisher (1934) proposed ANCOVA to improve efficiency: running OLS of Y_i on $(Z_i, Y_i, 1)$ and obtaining the coefficient of Z_i as an estimator for τ .
- ▶ Reanalysis of ANCOVA under the Neyman (1923)'s potential outcome framework.

6.2 Regression adjustment

► Freedman (2008b,a):

1. $\hat{\tau}_F$ is biased but the simple difference in means estimator $\hat{\tau}$ is unbiased.
2. The asymptotic variance of $\hat{\tau}_F$ may be even larger than that of $\hat{\tau}$ when $n_1 \neq n_0$.
3. The standard error from the OLS is inconsistent for the true standard error of $\hat{\tau}_F$ under the CRE.

► Lin (2013):

1. The bias of $\hat{\tau}_F$ is small in large samples and it goes to zero as $n \rightarrow \infty$.
2. We can improve the asymptotic efficiency of both $\hat{\tau}$ and $\hat{\tau}_F$ by using the coefficient of Z_i in the OLS of Y_i on $(1, Z_i, Y_i, Z_i \times Y_i)$. Let $\hat{\tau}_L$ be the proposed estimator by Lin (2013) then the EHW standard error is a conservative estimator for the true standard error of $\hat{\tau}_L$ under the CRE.
3. The EHW standard error for $\hat{\tau}_F$ in the OLS fit of Y_i on $(1, Z_i, Y_i)$ is a conservative estimator for the true standard error of $\hat{\tau}_F$ under the CRE.

Eicker–Huber–White Standard Error

Let $\{x_i, y_i\}_{i=1}^n$ be i.i.d., x_i be $k \times 1$ and y_i be a scalar. Let $\mathbf{X} = (x_1^T, \dots, x_n^T)^T$ ($n \times k$) and $\mathbf{Y} = (y_1, \dots, y_n)^T$ ($n \times 1$). Suppose a regression of $\mathbf{Y}$ on $\mathbf{X}$ ($\mathbf{Y} = \mathbf{X}^T \beta + \mathbf{e}$) then trivially, the coefficient of the regression $\hat{\beta}$ ($k \times 1$) is:

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} (\mathbf{X}^T \mathbf{Y}),$$

then

$$\text{var}(\hat{\tau}) = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{e}^T \mathbf{e} \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1}.$$

So the moment estimator for the variance of $\hat{\beta}$ is

$$\text{var}(\hat{\tau}) = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \hat{\mathbf{e}}^T \hat{\mathbf{e}} \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1},$$

which is **Eicker–Huber–White** robust variance estimator.

6.2 Regression adjustment

- ▶ Building on Neyman (1923), we can reduce the variance of the estimator $\hat{\tau}$ by reducing the variance of the potential outcomes.
- ▶ A family of linearly adjusted estimators is

$$\begin{aligned}\hat{\tau}(\beta_0, \beta_1) &= \frac{1}{n_1} \sum_{i=1}^n Z_i(Y_i - \beta_1^T X_i) - \frac{1}{n_0} \sum_{i=1}^n (1 - Z_i)(Y_i - \beta_0^T X_i) \\ &= \{\hat{\bar{Y}}(1) - \beta_1^T \hat{\bar{X}}(1)\} - \{\hat{\bar{Y}}(0) - \beta_0^T \hat{\bar{X}}(0)\},\end{aligned}$$

where $\{\hat{\bar{Y}}(0), \hat{\bar{Y}}(1)\}$ is the sample means of the outcomes and $\{\hat{\bar{X}}(0), \hat{\bar{X}}(1)\}$ is the sample means of the covariates.

- ▶ $\hat{\tau}(\beta_0, \beta_1)$ tries to reduce the variance of $\hat{\tau}$ by residualizing the potential outcomes. (Similar motivation to IV?)

6.2 Regression adjustment

- ▶ With $\beta_0 = \beta_1 = 0$, $\hat{\tau}(0, 0) = \hat{\tau}$.
- ▶ It has mean τ for any fixed values of β_0 and β_1 because $\bar{X} = 0$.
- ▶ We are interested in finding (β_0, β_1) s.t. minimizes $\text{var}(\hat{\tau}(\beta_0, \beta_1))$. Hence, this is the Neyman (1923)'s potential outcome framework with $\{\hat{Y}(1) - \beta_1 \hat{X}(1), \hat{Y}(0) - \beta_0 \hat{X}(0)\}$.
- ▶ Therefore, building on Neyman (1923), this estimator has a variance:

$$\text{var}(\hat{\tau}(\beta_0, \beta_1)) = \frac{S^2(1; \beta_1)}{n_1} + \frac{S^2(0; \beta_0)}{n_0} - \frac{S^2(\tau; \beta_1, \beta_0)}{n}.$$

6.2 Regression adjustment

- Similarly, we obtain the Neyman (1923)'s conservative variance:

$$\widehat{V}(\beta_0, \beta_1) = \frac{\widehat{S}^2(1; \beta_1)}{n_1} + \frac{\widehat{S}^2(0; \beta_0)}{n_0},$$

where

$$\begin{aligned}\widehat{S}^2(1; \beta_1) &= \frac{1}{n_1 - 1} \sum_{i=1}^n Z_i (Y_i - \gamma_1 - \beta_1^T X_i) \\ , \widehat{S}^2(0; \beta_0) &= \frac{1}{n_0 - 1} \sum_{i=1}^n (1 - Z_i) (Y_i - \gamma_0 - \beta_0^T X_i),\end{aligned}$$

with $\gamma_1 = \frac{1}{n_1} \sum_{i=1}^n Z_i (Y_i - \beta_1^T X_i)$ and $\gamma_0 = \frac{1}{n_0} \sum_{i=1}^n (1 - Z_i) (Y_i - \beta_0^T X_i)$.

6.2 Regression adjustment

To find (β_0, β_1) s.t. minimizes $\widehat{V}(\beta_0, \beta_1)$, we solve the following objects:

$$\min_{\gamma_1, \beta_1} Z_i(Y_i - \gamma_1 - \beta_1^T X_i)^2, \quad \min_{\gamma_0, \beta_0} (1 - Z_i)(Y_i - \gamma_0 - \beta_0^T X_i)^2.$$

Finally, we obtain the final estimator:

$$\widehat{\tau}(\widehat{\beta}_1, \widehat{\beta}_0) = \{\widehat{\bar{Y}}(1) - \widehat{\beta}_1^T \widehat{\bar{X}}(1)\} - \{\widehat{\bar{Y}}(0) - \widehat{\beta}_0^T \widehat{\bar{X}}(0)\}.$$

Obviously²,

$$\widehat{\bar{Y}}(1) = \widehat{\gamma}_1 + \widehat{\beta}_1^T \widehat{\bar{X}}(1), \quad \widehat{\bar{Y}}(0) = \widehat{\gamma}_0 + \widehat{\beta}_0^T \widehat{\bar{X}}(0). \quad (2)$$

Then we rewrite the estimator as:

$$\widehat{\tau}(\widehat{\beta}_1, \widehat{\beta}_0) = \widehat{\gamma}_1 - \widehat{\gamma}_0. \quad (3)$$

²The mean of residuals equals zero.

6.2 Regression adjustment

Proposition 6.2

The estimator $\hat{\tau}(\hat{\beta}_1, \hat{\beta}_2)$ in (3) equals the coefficient of Z_i in the OLS fit of Y_i on $(1, Z_i, X_i, Z_i \times X_i)$, which is Lin (2013)'s estimator $\hat{\tau}_L$ introduced before.

Proof.

Let $Y_i = a + bZ_i + c^T X_i + d^T(Z_i X_i) + e_i$ be regression of Y_i on $(1, Z_i, X_i, Z_i \times X_i)$. We have $Y_i = a + c^T X_i$ for control group ($Z_i = 0$) and $Y_i = a + b + (c + d)^T X_i$ for treatment group ($Z_i = 1$). Recall (2), then we obtain

$$a = \hat{\gamma}_0, \quad b = \hat{\beta}_1, \quad a + b = \hat{\gamma}_1, \quad c + d = \hat{\beta}_1.$$

Hence, $b = \hat{\gamma}_1 - \hat{\gamma}_0$. $\square$

6.2 Regression adjustment

The Neyman (1923)'s conservative variance estimator for $\hat{\tau}_L$ is

$$\hat{V}(\hat{\beta}_1, \hat{\beta}_2) = \frac{1}{n_1(n_1 - 1)} \sum_{i=1}^n Z_i (Y_i - \hat{\gamma}_1 - \hat{\beta}_1^T X_i)^2 + \frac{1}{n_0(n_0 - 1)} \sum_{i=1}^n (1 - Z_i) (Y_i - \hat{\gamma}_0 - \hat{\beta}_0^T X_i)^2.$$

- ▶ Lin (2013) further showed that the EHW standard error from the OLS in Prop. 24 is almost identical to $\hat{V}(\hat{\beta}_1, \hat{\beta}_2)$ which is a conservative estimator of the true standard error of $\hat{\tau}_L$ under the CRE.
- ▶ This is because we do not assume that the linear model is correctly specified and the EHW standard error is robust to model misspecification.

6.2 Regression adjustment

- ▶ The estimator $\hat{\tau}(\hat{\beta}_1, \hat{\tau}_2)$ uses two estimated coefficients then this uncertainty may cause finite-sample bias in the final estimator.
- ▶ Lin (2013) showed that this issue goes away asymptotically³.
- ▶ In finite samples, regression adjustment can be **even harmful** due to the additional uncertainty in the estimated coefficients $\hat{\beta}_1$ and $\hat{\beta}_0$.

³The asymptotic theory requires a large sample size and some regularity conditions on the potential outcomes and covariates.

6.2 Regression adjustment

- ▶ We can view Lin (2013)'s estimator as a *predictive estimator* based on OLS fits of the potential outcomes on the covariates.
- ▶ Using data from both groups:

$$\hat{\mu}_1(X) = \hat{\gamma}_1 + \hat{\beta}_1^T X, \quad \hat{\mu}_0(X) = \hat{\gamma}_0 + \hat{\beta}_0^T X.$$

- ▶ Ideally, we want to estimate the average causal effect $\tau = n^{-1} \sum_{i=1}^n (Y_i(1) - Y_i(0))$ but we cannot observe both $Y_i(1), Y_i(0)$ jointly.
- ▶ Here, we use the prediction models then we obtain the following predictive estimator:

$$\hat{\tau}_{\text{pred}} = \frac{1}{n} \left\{ \sum_{Z_i=1} Y_i + \sum_{Z_i=0} \hat{\mu}_1(X_i) - \sum_{Z_i=1} \hat{\mu}_0(X_i) - \sum_{Z_i=0} Y_i \right\}.$$

- ▶ $\hat{\tau}_{\text{pred}} = \hat{\tau}_L$.

6.2 Regression adjustment

- ▶ If we predict all the potential outcomes, then we have the following *projective estimator*:

$$\hat{\tau}_{\text{proj}} = \frac{1}{n} \sum_{i=1}^n (\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)).$$

- ▶ $\hat{\tau}_{\text{proj}} = \hat{\tau}_L$
Proof.

$$\begin{aligned} \hat{\tau}_{\text{proj}} &= \frac{1}{n} \sum_{i=1}^n (\hat{\gamma}_1 + \hat{\beta}_1^T X_i - \gamma_0 - \hat{\beta}_0^T X_i) = \frac{1}{n} \sum_{i=1}^n (\hat{\gamma}_1 - \hat{\gamma}_0) + (\hat{\beta}_1 - \hat{\beta}_0)^T \frac{1}{n} \sum_{i=1}^n X_i \\ &= \hat{\gamma}_1 - \hat{\gamma}_0 = \hat{\tau}_L. \quad (\because \frac{1}{n} \sum_{i=1}^n X_i = 0) \quad \square \end{aligned}$$

- ▶ Here, we focus on linear projection but the predictor does not always have to be a linear predictor.

6.2 Regression adjustment

The linearly adjusted estimator has an equivalent form⁴

$$\hat{\tau}(\beta_1, \beta_0) = \hat{\tau} - \gamma^T \hat{\tau}_X, \quad (4)$$

where $\gamma = \frac{n_0}{n}\beta_1 + \frac{n_1}{n}\beta_0$, so we can also rewrite it as $\hat{\tau}(\gamma) = \hat{\tau}(\beta_1, \beta_0)$. Similarly, Lin (2013)'s estimator has an equivalent form

$$\hat{\tau}_L = \hat{\tau} - \hat{\gamma}^T \hat{\tau}_X. \quad (5)$$

► Proof

⁴ $\hat{\tau}$ and $\hat{\tau}_X$ are the difference-in-means estimator for outcomes and covariates, respectively.

6.2 Regression adjustment

- ▶ Eq. (4) and (5) essentially subtract some linear combinations of the difference in means of the covariates.
- ▶ Since $\hat{\tau}$ and $\hat{\tau}_X$ are correlated under the CRE (in general), the covariate adjustment with an appropriately chosen γ reduces the variance of $\hat{\tau}$.
- ▶ The final estimators depend only on γ or $\hat{\gamma}$, so the choice of the β -coefficients is **not unique**.

6.2 Regression adjustment

- ▶ Rerandomization and Lin (2013)'s regression adjustment are duals.
- ▶ When a is small, the asymptotic distribution of $\hat{\tau}$ under ReM is almost identical to the asymptotic distribution of $\hat{\tau}_L$ under the CRE.
- ▶ $\rightsquigarrow$ ReM uses covariates in the *design stages* and Lin (2013)'s regression adjustment uses covariates in the *analysis stages*.

6.2 Regression adjustment

If we have discrete covariate C_i with K covariates, we can create $K - 1$ centered dummy variables

$$X_i = (I(C_i = 1) - \pi_{[1]}, \dots, I(C_i = K - 1) - \pi_{[K-1]}),$$

where $\pi_{[k]}$ is the proportion of unites with $C_i = k$. In this case, Lin (2013)'s regression adjustment is equivalent to post-stratification.

Proposition 6.3

$\hat{\tau}_L$ based in X_i is numerically identical to the post-stratification estimator $\hat{\tau}_{PS}$ based on C_i .

► Proof

6.2 Regression adjustment

- ▶ An important covariate X in many studies is the lagged outcome before the treatment
 - ▶ The covariate X is the pre-test score if the outcome Y is the post-test score in educational research
 - ▶ The covariate X is the log wage before the job training program if the outcome Y is the log wage after the job training program.
- ▶ With lagged covariates, a popular estimator is the *gain score* or *difference-in-difference* estimator with $\beta_1 = \beta_0 = 0$ in $\hat{\tau}(\beta_1, \beta_0)$:

$$\begin{aligned}\hat{\tau}(1, 1) &= \frac{1}{n_1} \sum_{i=1}^n Z_i(Y_i - X_i) - \frac{1}{n_0} \sum_{i=1}^n (1 - Z_i)(Y_i - X_i) \\ &= \left\{ \hat{Y}(1) - \hat{Y}(0) \right\} - \left\{ \hat{X}(1) - \hat{X}(0) \right\}.\end{aligned}$$

6.2 Regression adjustment

- ▶ $\hat{\tau}(1, 1)$ is unbiased with a conservative variance estimator:

$$\hat{V}(1, 1) = \frac{1}{n_1(n_1 - 1)} \sum_{i=1}^n Z_i \{g_i - \hat{g}(1)\}^2 + \frac{1}{n_0(n_0 - 1)} \sum_{i=1}^n (1 - Z_i) \{g_i - \hat{g}(0)\}^2,$$

with $g_i = Y_i - X_i$ and sample means of the gain score $\hat{g}(0), \hat{g}(1)$.

- ▶ Theoretically, Lin (2013)'s estimator is always more efficient but suffers from finite sample bias while $\hat{\tau}(1, 1)$ is unbiased.

6.2 Regression adjustment

- ▶ Suppose an experiment stratified on a discrete variable C and we also observe additional covariates X .
- ▶ We can extend Lin (2013)'s estimator to the SRE.
- ▶ Let $\hat{\tau}_{L,[k]}$ be Lin (2013)'s estimator within strata:

$$\hat{\tau}_{L,S} = \sum_{k=1}^K \pi_{[k]} \hat{\tau}_{L,[k]}.$$

- ▶ A conservative variance estimator is

$$\hat{V}_{L,S} = \sum_{k=1}^K \pi_{[k]}^2 \hat{V}_{\text{EHW},[k]}.$$

6.3 Unification, combination and comparison

- ▶ Li and Ding (2020) unified the literature and showed that we can combine rerandomization and regression adjustment.
- ▶ If we rerandomize in the design stage, we can use Lin (2013)'s estimator with the EHW standard error in the analysis stage.
- ▶ It improves covariate balance in the design stage and estimation efficiency in the analysis stage.

Proof of Eq. (1)

Neyman (1923) derives

$$\text{var}(\hat{\tau}) = \frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n}.$$

In this case, we replace $\hat{\tau}$ with $\hat{\tau}_X$ and because $X_i(0) = X_i(1) = X_i$, then $S_X^2(1) = S_X^2(0) = S_X^2$ and $S_X^2(\tau) = S_X^2\{X(1) - X(0)\} = 0$. Hence,

$$\text{cov}(\hat{\tau}_X) = \frac{1}{n_1} S_X^2 + \frac{1}{n_0} S_X^2. \quad \square$$

Proof of Lemma 7

Let $X_i^* = b_0 + BX_i$ then

$$\begin{aligned}\hat{\tau}_{X^*} &= \frac{1}{n_1} \sum_{i=1}^n Z_i X_i^* - \frac{1}{n_0} \sum_{i=1}^n (1 - Z_i) X_i^* \\ &= \left\{ \frac{1}{n_1} \sum_{i=1}^n Z_i b_0 - \frac{1}{n_0} \sum_{i=1}^n (1 - Z_i) b_0 \right\} + \left\{ \frac{1}{n_1} \sum_{i=1}^n Z_i B X_i - \frac{1}{n_0} \sum_{i=1}^n (1 - Z_i) B X_i \right\} \\ &= B \hat{\tau}_X.\end{aligned}$$

Substituting $\hat{\tau}_{X^*} = B \hat{\tau}_X$,

$$\text{cov}(\hat{\tau}_{X^*}) = \text{cov}(B \hat{\tau}_X) = B \text{cov}(\hat{\tau}_X) B^T.$$

Because B is invertible,

$$M^* = \hat{\tau}_{X^*}^T \text{cov}(\hat{\tau}_{X^*})^{-1} \hat{\tau}_{X^*} = \hat{\tau}_X^T B^T (B^T)^{-1} \text{cov}(\hat{\tau}_X)^{-1} B^{-1} B \hat{\tau}_X = M. \quad \square$$

Proof for Eq. (4) and (5)

Recall

$$\hat{\tau}(\beta_1, \beta_0) = \left\{ \hat{Y}(1) - \beta_1^T \hat{X}(1) \right\} - \left\{ \hat{Y}(0) - \beta_0^T \hat{X}(0) \right\}, \quad \hat{\tau}_X = \hat{X}(1) - \hat{X}(0).$$

By definition,

$$\hat{\tau}(\beta_1, \beta_0) = \left\{ \hat{Y}(1) - \hat{Y}(0) \right\} - \left\{ \beta_1^T \hat{X}(1) - \beta_0^T \hat{X}(0) \right\} = \hat{\tau} - \left\{ \beta_1^T \hat{X}(1) - \beta_0^T \hat{X}(0) \right\}.$$

Hence, we verify $\beta_1^T \hat{X}(1) - \beta_0^T \hat{X}(0) = \gamma^T \hat{\tau}_X$. Here, $\bar{X} = 0$, then $\frac{n_1}{n} \hat{X}(1) + \frac{n_0}{n} \hat{X}(0) = 0$.

Substituting $\gamma = \frac{n_0}{n} \beta_1 + \frac{n_1}{n} \beta_0$,

$$\begin{aligned} \gamma^T \hat{\tau}_X &= \frac{n_0}{n} \beta_1^T \hat{X}(1) - \frac{n_0}{n} \beta_1^T \hat{X}(0) + \frac{n_1}{n} \beta_0^T \hat{X}(1) - \frac{n_1}{n} \beta_0^T \hat{X}(0) \\ &= \beta_1^T \hat{X}(1) - \beta_0^T \hat{X}(0). \quad \square \end{aligned}$$

We similarly obtain Eq. (5). $\square$

Proof for Prop. 6.3

Recall the projective estimator

$$\hat{\tau}_L = \frac{1}{n} \sum_{i=1}^n \{\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)\}.$$

By definition, $\hat{\mu}_1(X_i) = \hat{\mu}_1(C_i)$, then

$$\hat{\tau}_L = \frac{1}{n} \sum_{k=1}^K \sum_{i; C_i=k} \{\hat{\mu}_1(k) - \hat{\mu}_0(k)\} = \sum_{k=1}^K \frac{n_{[k]}}{n} \{\hat{\mu}_1(k) - \hat{\mu}_0(k)\} = \sum_{k=1}^K \pi_{[k]} \hat{\tau}_{[k]} = \hat{\tau}_{PS}. \quad \square$$

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